

Aspirin: A History

The therapeutic use of salicylate-bearing plants substantially predates the isolation of any active compound. Preparations derived from willow bark are recorded in Sumerian clay tablets and in the Egyptian medical papyri, and the practice was sufficiently established in the classical world that references to willow decoctions appear in the Hippocratic corpus. These traditions persisted through the medieval period, though the empirical basis for the remedy remained obscure and its use was inconsistent.

A more systematic investigation began in eighteenth-century England. Edward Stone, a clergyman in Oxfordshire, reasoned by analogy from the bitterness of willow bark to that of cinchona, then valued as a febrifuge, and conducted an extended informal trial among his parishioners. His account, communicated to the Royal Society in 1763, described willow bark as an effective and considerably cheaper alternative to imported cinchona. Stone's reasoning rested on a doctrine of signatures that later chemistry would not vindicate, but his empirical conclusion proved durable.

The nineteenth century brought the isolation and characterisation of the responsible molecules. Salicin was obtained from willow, and salicylic acid was subsequently extracted from meadowsweet, whose Latin genus name would later supply a syllable to the eventual trade name. Sodium salicylate entered clinical use for rheumatic conditions, but its severe gastric irritation and unpalatability limited its acceptance. In 1853 Charles Frederic Gerhardt prepared acetylsalicylic acid by treating sodium salicylate with acetyl chloride, though the product was impure and Gerhardt did not pursue the reaction further.

The compound's transformation into a commercial pharmaceutical occurred at the Bayer company in Bavaria in 1897. Attribution of the achievement remains contested. The conventional account credits Felix Hoffmann with developing a synthesis yielding a stable, pure product; a competing account, advanced by Arthur Eichengrün in 1949, claims that Hoffmann worked under Eichengrün's direction and that Eichengrün's contribution was expunged from company records during the Nazi period on account of his Jewish ancestry. The historical evidence has been argued in both directions and no consensus has emerged.

Bayer marketed the drug internationally from 1899. Its subsequent commercial history was shaped as much by trademark law as by pharmacology. Under the terms of the Treaty of Versailles, Bayer forfeited its trademark rights in several Allied countries, and in the United States courts subsequently held that the term had become generic through the company's own failure to police its use. Aspirin accordingly remains a protected mark in Canada, Germany and numerous other jurisdictions while functioning as an ordinary common noun in the United States and Britain.

Interest in the drug declined following the introduction of paracetamol and ibuprofen in the mid-twentieth century, and might have continued to decline had John Vane not identified prostaglandin inhibition as the underlying mechanism in 1971, work for which he shared the Nobel Prize in 1982. The elucidation of the antiplatelet effect prompted a second commercial revival, and

low-dose aspirin became one of the most widely prescribed preventive therapies in cardiovascular medicine.